Ganita Manjari — Grade 9 (Part II)

Chapter 14: Math of Space (Surface Area and Volume) — Exercise Set 14.2 Solutions

Question 1

> **Two cylinders, A and B, are given. The radius of cylinder B is twice that of cylinder A, and the height of cylinder B is half that of cylinder A. Find the ratio of the curved surface area of A to the curved surface area of B. Also find the ratio of the volume of A to the volume of B.**

Easy Step-by-Step Solution:

Let radius of cylinder A = r, height of cylinder A = h.

Then radius of cylinder B = $2r$, height of cylinder B = $h / 2$.

Curved Surface Area (CSA) = $2\pi rh$ | Volume (V) = $\pi r^2 h$

Part 1: Ratio of CSA

CSA(A) = $2\pi rh$

CSA(B) = $2\pi(2r)(h/2) = 2\pi rh$

Ratio of CSA = $CSA(A) / CSA(B) = 2\pi rh / 2\pi rh = 1 / 1 = 1 : 1$

Part 2: Ratio of Volume

V(A) = $\pi r^2 h$

V(B) = $\pi(2r)^2(h/2) = \pi(4r^2)(h/2) = 2\pi r^2 h$

Ratio of Volume = $V(A) / V(B) = (\pi r^2 h) / (2\pi r^2 h) = 1 / 2 = 1 : 2$

Final Answers: Ratio of CSA = 1 : 1 | Ratio of Volume = 1 : 2

Question 2

The radii of two cylinders are in the ratio 2:3, and their heights are in the ratio 3:2. Find (a) the ratio of their volumes, and (b) the ratio of their curved surface areas.

Easy Step-by-Step Solution:

Let $r_1/r_2 = 2/3$ and $h_1/h_2 = 3/2$.

(a) Ratio of Volumes:

$V_1 / V_2 = (\pi r_1^2 h_1) / (\pi r_2^2 h_2) = (r_1/r_2)^2 \times (h_1/h_2)$

$V_1 / V_2 = (2/3)^2 \times (3/2) = (4/9) \times (3/2) = 12 / 18 = 2 / 3$

(b) Ratio of Curved Surface Areas:

$CSA_1 / CSA_2 = (2\pi r_1 h_1) / (2\pi r_2 h_2) = (r_1/r_2) \times (h_1/h_2)$

$CSA_1 / CSA_2 = (2/3) \times (3/2) = 1 / 1$

Final Answers: (a) 2 : 3 | (b) 1 : 1

Question 3

The edge of a cube measures r cm. The largest possible right circular cylinder is cut out of the cube. What do you think is the volume of the cylinder (in cm³)?

Easy Step-by-Step Solution:

For the largest cylinder inside a cube of side r:

• Height of cylinder (h) = r

• Diameter of cylinder base = $r \Rightarrow Radius\ (R) = r / 2$

Volume of cylinder:

$V = \pi R^2 h = \pi (r/2)^2 (r) = \pi (r^2/4) r = (\pi r^3) / 4$

Final Answer: (π r³) / 4 cm³

Question 4

The radius of the base of a cylinder is increased by 10%. At the same time, the height of the cylinder is decreased by x%. Given that the volume of the cylinder remains unchanged, find the value of x.

Easy Step-by-Step Solution:

Let initial radius = r, height = h.

New radius $r' = 1.10\, r$

New height $h' = h\,(1 - x/100)$

Since volume remains unchanged: $\pi\,(r')^2\,h' = \pi\, r^2\, h$

$(1.10\, r)^2 \times h' = r^2\, h$

$1.21\, r^2\, h' = r^2\, h \Rightarrow h' = h / 1.21 = h / (121/100) = 100/121\, h$

Decrease fraction in height = $1 - 100/121 = 21/121$

Percentage decrease $x = (21 / 121) \times 100\ \% = 2100 / 121\ \% \approx 17.355\%$

Final Answer: x = 2100/121 ≈ 17.36%

Question 5

A solid metallic cube of side 12 cm is melted and recast into solid cylindrical rods, each having radius 2 cm and height 12 cm. Find:
(i) the volume of the cube,
(ii) the volume of one cylindrical rod,
(iii) the approximate number of complete cylindrical rods that can be formed (π ≈ 22/7).

Easy Step-by-Step Solution:

(i) Volume of the cube:

$$V_{cube} = side^3 = 12^3 = 1728\ cm^3$$

(ii) Volume of one cylindrical rod:

$$V_{rod} = \pi r^2 h = (22/7) \times (2)^2 \times 12 = (22/7) \times 4 \times 12 = 1056 / 7 \approx 150.857\ cm^3$$

(iii) Approximate number of complete rods:

$$Number\ of\ rods = V_{cube} / V_{rod} = 1728 / (1056 / 7) = (1728 \times 7) / 1056 = 12096 / 1056 = 11.45$$

Since only *complete* rods can be formed, we take the integer part: **11 complete rods**.

Final Answers: (i) 1728 cm³ | (ii) 1056/7 cm³ (≈ 150.86 cm³) | (iii) 11 complete rods